



CHEM Project

**Advanced Decision Support System for
Chemical/Petrochemical Manufacturing Processes**

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iFuzzyFDI Manual

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1. Introduction

The aim of the iFUZZYFDI is to determine in on-line mode the *faults* existing in the monitored process (*fault detection and isolation*). It is done by calculating *fault certainty factors*, or simply *fault factors* on the basis of analysis the *residuals values*. This task is done automatically in two steps:

- ◆ First, residual evaluation is carried out - *diagnostic signals* are calculated which are quantitative representation of residuals. This is computed comparing the residual values with predefined threshold either in a crisp or fuzzy way. Exceeding the threshold values by any of the residuals indicates the existence of fault in the system.
- ◆ Then, reasoning about fault existing in the system is carried out basing on observer *symptoms* (symptom is such a value of diagnostic signal that testify about abnormal system behaviour) and according to fuzzy logic reasoning rules - comparison of the current pattern of the diagnostic signals values with the reference ones.

To enable fulfilling the above mentioned tasks, the iFUZZYFDI provides a GUI for setting its configuration parameters. This GUI allows:

- ◆ Setting residuals evaluation parameters. Basing on statistical parameters of the residuals values or expert knowledge, the user defines the parameters of fuzzy sets that describe qualitative representation of residuals values (i.e., diagnostic signals).
- ◆ Basing on expert knowledge, the user defines *faults-symptoms-time relation* used for diagnostic reasoning. This relation describes time relation between observer symptoms (i.e., diagnostic signal values) and faults appearing in the system.

2. Running the iFUZZYFDI - step by step

Using of the iFUZZYFDI can be divided into phases. Firstly, the user must configure the parameters with use of configuration module FISCONFIGURATOR. Then, one can start fault detection and isolation procedures. They are realised by on-line calculation module RTFAULTISOLAOTR (real time).

The user must perform several steps to configure the iFUZZYFDI. All needed operations are listed below. They are presented in most suitable to do order:

- ◆ **Define non-model based residuals.** Model based residuals are defined during MITFORRD RT SIMULATOR configuration. Here, the user can add some non-model based ones.
- ◆ **Default considered faults.** The user must define the set of residuals that the system should monitor.
- ◆ **Configure residual evaluation parameters.** One of the most important think is to set up residual evaluation parameters. This step is repeated several times until maximal possible fault detectability and distinguishability is achieved.
- ◆ **Define GlobalFISes.** The user must group faults and residuals into separated groups – Global FISes. This groups usually correspond to separate sections (divisions) of the process. The FDI procedures are realised independently for the groups separated on this level. For not complex processes one Global FIS covering all residuals and faults is usually enough.
- ◆ **Define diagnostic relation.** For each Global FIS the user must define diagnostic relation describing the relation between faults appearing in the system and the symptoms that are observed. This is the most important knowledge the user must introduce into the configuration.
- ◆ **Analyse FDI report on GlobalFIS level.** After defining diagnostic relation one can evaluate achieved fault detectability and isolability. It can be done by analysing FDI report. Usually, the user comes to the conclusion that some configuration parameters can be changed to achieve better fault distinguishability. The steps 3 and 5 are repeated several times until the best possible solution is achieved.

- ◆ **Define SubFISes.** The division into SubFISes is made taking into account diagnostic relation and FDI report results. Division on this level is made mainly due to:
 - ◆ decrease sensitivity for wrong residual evaluation, the consequences of observing improper symptoms (*wrong residual evaluation e.g. due to bigger modelling errors, noises, etc*) are limited to only one subsystem,
 - ◆ increase possibility of detecting multi faults, multi faults erased in different subsystems are easily detected and isolated.

There exist a connection of the information elaborated on the SubFISes level. This information is accommodated on the GlobalFIS level. The user received the information form the top level of reasoning.

- ◆ **Analyse FDI report on SubFIS level.** One can evaluate achieved fault detectability and isolability on SubFIS level. Somotemis steps 3, 5 or 7 are repeated until the best possible solution is achieved.
- ◆ **Adjust FDI algorithm parameters.** The system automatically choose the default FDI algorithm parameters. In most cases they are good enough. However, the user have a possibility to change this parameters, both on the GobalFIS and SubFIS levels.
- ◆ **Start the RTFaultIsolator.** After finalising setting of configuration parameters the iFUZZYFDI is ready to start. The user must only launch the module. The subscription to the necessary topics, under which the residuals are published, will be done automatically.

The knowledge about the process become deeper and deeper during the time. Sometimes, the new knowledge is introduced to the system, several above mentioned steps are repeated. Therefore, the iFUZZYFDI was design to enable easy reconfiguration. All above mentioned tasks can be done with use of specialised GUI's described in section 5.

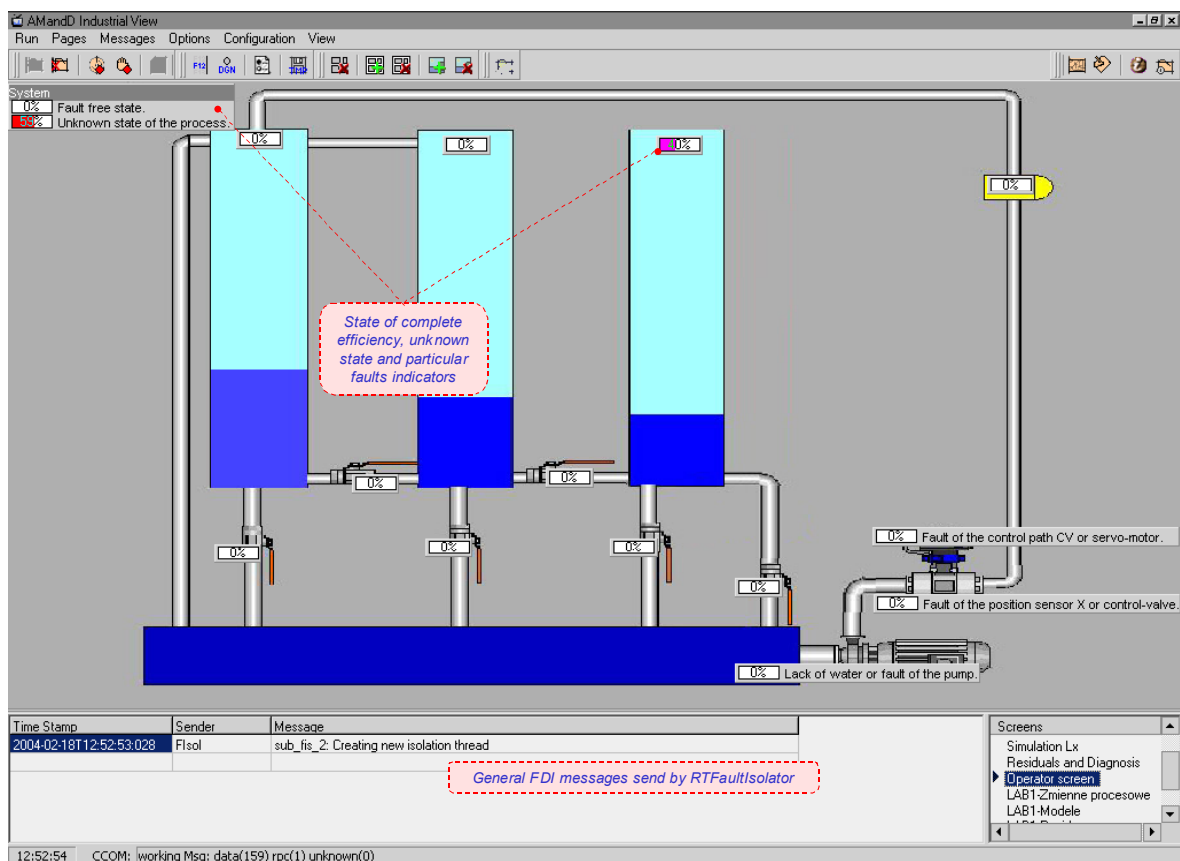


Figure 1. Visualisation of elaborated diagnosis about faults in the system.

During on-line calculations the RTFAULTISOLATOR can work completely automatically. All the calculation are done automatically after receiving new residual values. In each step of calculations, the module calculates certainty factors for all considered (defined during configuration and

enabled) fault. The certainty factor equal 0 means that the system is completely sure that this fault does not exist. If the certainty factor equals 1 it means that the system is completely sure that the fault exist in the system. Elaborated certainty factors are published under defined topic. The utilisation of achiever results depends on the user and system configuration. The exemplary use is shown on Figure 1. In this case fault certainty factors are shown to the operator with use of visualisation module INVIEW.

There are few configuration parameters which can be changed by the user in on-line mode. They are as follows:

- ◆ **Enable / disable residuals.** The user can temporary switch on and off the residuals from calculations.
- ◆ **Enable / disable faults.** The user can temporary switch on and off the faults that are monitored.
- ◆ **Set fault as present.** The user can inform the system about existing faults by setting their status to 'present'. Such faults are extruded from reasoning, their certainty factors are automatically set to 1. Setting present fault will trigger automatic reconfiguratin.
- ◆ **Close isolation thread.** The user can close one or all isolation thread. It triggers refreshing calculations if the symptoms are still observed.
- ◆ **Change isolatio mode.** The isolation mode of each existing isolation thread can be change for the user request.
- ◆ **Acknowledge isolation thread result.** The user must acknowledge the elaborated information about fault existing in the system to trigger automatic reconfiguration. It can be done automatically only under some specific conditions (when final, not contradictory and certain diagnosis is elaborated).

All this tasks can be done through GUI of the RTFaultIsolator and by calling proper RPC functions. They are described in section 5 (GUI) and section 4.1 (RPC).

3. Input and Output data

The Figure 2 shows the main iFUZZYFDI activities A0 and C0 and their main ICOMs.

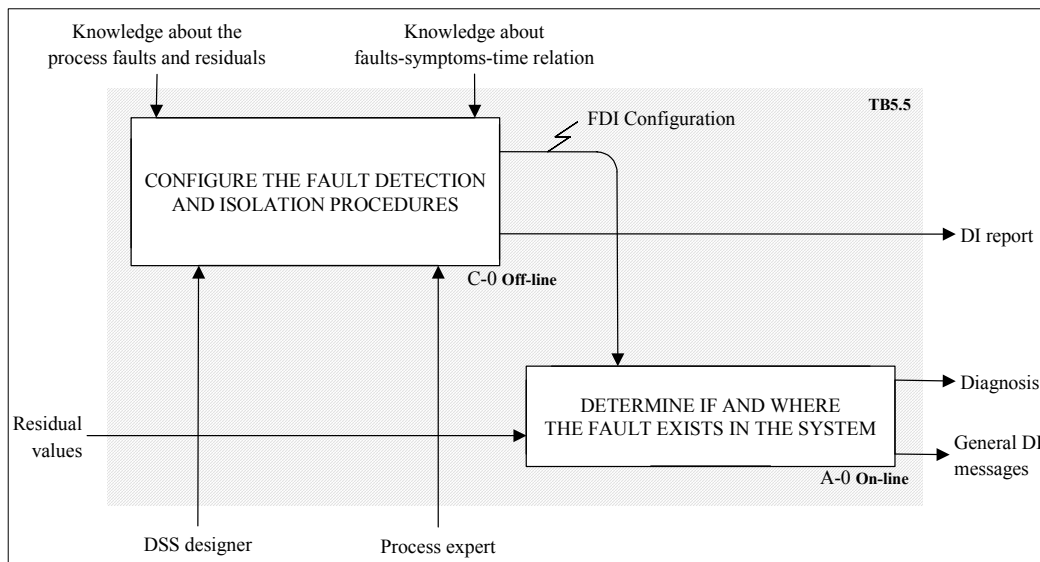


Figure 2. Activities of the iFuzzyFDI and their main ICOMs

3.1. Inputs

- ◆ **Knowledge about the process faults and residuals.** The knowledge about process faults and residuals represents all the information about considered faults and available residuals, *i.e.*, the definition of the set of faults and residuals, the parameters of residuals, behavior of residuals in fault free state, etc. The user can use residuals defined by other AMANDD system modules or add new ones. The configuration of faults corresponds to system structure and its analysis. This toolbox input is used in off-line mode during configuration stage.
- ◆ **Knowledge about the faults-symptoms-time relation.** The knowledge about *diagnostic relation* (faults-symptoms-time relation) is a kind of *diagnostic model*. This relation defines the pattern values of diagnostic signals, which should occur when the fault appears in the system. It can additionally take into account the symptoms dynamics. This knowledge can be extracted from different sources, *e.g.*, expert knowledge, structural, behavior or functional models, bound graphs, etc, and passed by the user to the toolbox by specially designed GUI. This toolbox input is used in off-line mode during configuration stage.
- ◆ **Residual values.** Current values of residuals are delivered by other modules with use of publish-subscribe mechanism. Delivering a new values of residuals triggers toolbox calculations. As a result new elaborated diagnosis (values of fault certainty factors) is produced as toolbox output. This toolbox input is used in on-line mode during monitoring of the system state. Only messages with “Residual” subject are decoded.

```
<CHENMessageContent>
  <Subject>Residuals</Subject>
  <DRIFT_DETECTION_ANALYSIS ID="residual_21">
    <Quantity Magnitude="-1.3512"/>
    <Time_Stamp>2004-02-14T15:14:58:549</Time_Stamp>
  </DRIFT_DETECTION_ANALYSIS>
  <DRIFT_DETECTION_ANALYSIS ID="residual_17">
    <Quantity Magnitude="-0.395119"/>
    <Time_Stamp>2004-02-14T15:14:58:549</Time_Stamp>
  </DRIFT_DETECTION_ANALYSIS>
</CHENMessageContent>
```

Figure 3. Example of residual CCOM message

3.2. Outputs

- ◆ **Diagnosis.** The diagnosis is the worked out information about the process faults. It is provided as a set of pairs <fault ID, fault certainty factor>. This toolbox input is used in on-line mode during monitoring of the system state. The subject of the message is “Faults”.

```
<
  <
    >Faults</
  <
    <
      <
        >0.15"/>
      <
        >2004-02-14T15:30:36:042</
      </
    <
      <
        >0"/>
      <
        >2004-02-14T15:30:36:042</
      </
    <
      <
        >0.85"/>
      <
        >2004-02-14T15:30:36:042</
      </
    </
  </
  >
```

Figure 4. Example of diagnosis CCOM message

- ◆ **General FDI messages.** The toolbox sends the special messages about creating new isolation threads (fault detection), reconfiguration, closing isolation threads, etc. Format of the message is shown on Figure 5. This toolbox input is used in on-line mode during monitoring of the system state.

Table 1. The available FDI Messages content

Message Content	Meaning
<i>Creating new isolation thread</i>	Information about controlling the isolation thread: creating, closing, reconfiguring
<i>Closing isolation thread</i>	
<i>Closing isolation thread for user request</i>	
<i>Combining existing isolation thread with new one</i>	
<i>Change to Multi-fault safe mode</i>	Information about changing the isolation mode.
<i>Change to only symptoms mode</i>	
<i>Change to cumulative symptoms time mode</i>	
<i>Change to BPR mode</i>	
<i>Change to RT mode</i>	
<i>Fault decay detected</i>	Information about detection of fault decay
<i>Final diagnosis elaborated</i>	Information about diagnosis elaboration
<i>Inconsistent diagnosis elaborated</i>	
<i>Unexpected symptom due to diagnostic relation</i>	Information about detection of abnormal symptoms behavior
<i>Time of detected inconsistent with time relation</i>	
<i>Symptom decay found</i>	

```

<CHERMMessageContent>
  <Subject>FDIMessages</Subject>
  <FDI_Message_SubFIS_ID="sub_fis_2" ThreadFIS_ID="thread_fis_1">
    <Message_Content> Creating new isolation thread </Message_Content>
    <Time_Stamp>2004-02-14T15:14:58:549</Time_Stamp>
  </FDI_Message>
</CHERMMessageContent>

```

Figure 5. Structure of FDI Message send by iFUZZYFDI

- ♦ **DI report.** DI report includes the information about achieved detectability and isolability of the considered faults in respect to defined faults-symptoms-time relation. It takes into account the faults, residuals and diagnostic rules statuses, reconfiguration options, etc. DI report is available in one of the GUI of iFUZZYFDI FIS CONFIGURATOR. It can be also exported (saved) in XML form. This toolbox output is used in off-line mode.

4. Toolbox Interfaces

4.1. Available RPCs

There are several RPC calls defined to control the behaviour of on-line calculation module RT FAULT ISOLATOR. They can be divided into three groups of RPC's that:

- ♦ control the behaviour of the whole module
- ♦ change the configuration parameters
- ♦ control the isolation threads

All RPC's follow the standard defined in CHEM platform. Table 2 presents the list of RPC's, their syntax and meaning.

```

<FDI_Report>
  <Present_Faults>
    <Fault ID="fault_2" name="(F3,F4)" />
  </Present_Faults>
  <Isolable_Faults>
    <Fault ID="fault_1" name="(F1,F2)" />
    <Fault ID="fault_3" name="F7" />
    <Fault ID="fault_6" name="F9" />
    <Fault ID="fault_11" name="F14" />
    <Fault ID="fault_12" name="F15" />
    <Fault ID="fault_13" name="F16" />
  </Isolable_Faults>
  <Conditionally_Isolable_Faults>
    <Fault_Group ID="fault_9" name="F12">
      <Fault ID="fault_5" name="F8" />
    </Fault_Group>
    <Fault_Group ID="fault_10" name="F13">
      <Fault ID="fault_8" name="F11" />
    </Fault_Group>
    <Fault_Group ID="fault_5" name="F8">
      <Fault ID="fault_7" name="F10" />
      <Fault ID="fault_9" name="F12" />
    </Fault_Group>
    <Fault_Group ID="fault_7" name="F10">
      <Fault ID="fault_5" name="F8" />
    </Fault_Group>
    <Fault_Group ID="fault_8" name="F11">
      <Fault ID="fault_10" name="F13" />
    </Fault_Group>
  </Conditionally_Isolable_Faults>
  <Unisolable_Faults>
    <Fault_Group ID="fault_7" name="F10">
      <Fault ID="fault_9" name="F12" />
    </Fault_Group>
  </Unisolable_Faults>
  <Undetectable_Faults>
    <Fault ID="fault_3" name="(F5,F6)" />
  </Undetectable_Faults>
  <Suspended_Diagnostic_Signals>
    <Diagnostic_Signal ID="residual_1" name="R1" />
  </Suspended_Diagnostic_Signals>
</FDI_Report>

```

Figure 6. Structure of the DI report in XML form

Table 2. The available FDI Messages content

Type	RPC name	RPC Parameters		
		#	Type	Value / Description
control RTFAULTISOLATOR	<i>Suspend</i>	Suspends calculations		
	<i>Resume</i>	Resume calculations		
	<i>Stop</i>	Stop calculations and exit the program		
	<i>Reconfigure</i>	Suspend calculations, reconfigure and resume calculations		
change configuration	<i>SetResidualStatus</i>	1	String	residual ID
		2	String	enabled / disabled
	<i>SetFaultStatus</i>	1	String	fault ID
		2	String	enabled / disabled
	<i>SetFaultPresence</i>	1	String	fault ID
		2	String	true / false
	<i>GetFullStatus</i>	Returns full description of RTFAULTISOLATOR status		
control FIThreads	<i>SetThreadMode</i>	1	String	isolation thread ID
		2	String	mode: BPR, OS_STD, OS_MFSAVE, CST
	<i>CloseThread</i>	1	String	isolation thread ID
	<i>CloseAllThreads</i>	Close all existing isolation thread (thread FISes)		
	<i>AcknowledgeThread</i>	1	String	isolation thread ID


```

<
  <
    >
    <
      ID="Main_FIS(global_fis_2)" status="at least one infinite isolation thread">
        let
          value="0"/>
        <
          <
            ID="F15(fault_37)" isolated="false" present="false">
              <
                quantity="1" status="unknown" time_stamp="2004-02-14T15:36:27:093"/>
              </>
            </>
          </>
        <
          ID="FIS_2(sub_fis_2)" status="at least one infinite isolation thread">
            let
              value="0"/>
            <
              <
                ID="F15(fault_37)" isolated="false">
                  <
                    quantity="1" status="unknown" time_stamp="2004-02-14T15:36:27:093"/>
                  </>
                </>
              </>
            </>
          </>
        <
          ID="global_fis_2_sub_fis_2_tFIS_1" initiation_time="2004-02-15T15:29:40:880"
            at_time="2004-02-15T15:29:40:880"
            initiator="R5(residual_20)" maximum_isolation_time="inf" mode="BPR"
            status="infinite">
            <
              ID="R3(residual_18)" cumulative_symptom_time="inf"
                cumulative_time="inf"
                determined="false" is_detection="true"/>
            </>
            <
              ID="F15(fault_37)" rejected="false"/>
            </>
            <
              status="temporary">
            </>
            <
              ID="F15(fault_37)" isolated="false">
                <
                  quantity="1" status="unknown"
                    time_stamp="2004-02-14T15:36:27:093"/>
                </>
              </>
            </>
          </>
        <
          <
            fault_id="F15(fault_37)">
            <
              <
                diagnostic_signal_id="R3(residual_18)">
                <
                  max="0" min="0"/>
                </>
              </>
              <
                diagnostic_signal_id="R4(residual_19)">
                <
                  max="0" min="0"/>
                </>
              </>
              <
                diagnostic_signal_id="R5(residual_20)">
                <
                  max="0" min="0"/>
                </>
              </>
              <
                diagnostic_signal_id="R6(residual_21)">
                <
                  max="inf" min="inf"/>
                </>
              </>
            </>
          </>
        </>
      </>
    </>
  </>
</>

```

Figure 7. Structure of the response for *GetFullStatus* RPC call

Table 3. Explanations of the response for *GetFullStatus* RPC call

Section / Sub-elements	Description
GLOBAL_FIS	Each GlobalFIS defined during configuration has one section describing its status. GlobalFISes are identified by unique ID. ID is displayed next to GlobalFIS name (in brackets). Status is a general description of GlobalFIS.
STATE_OF_COMPLETE_EFFICIENCY	Is a certainty factor of the state of complete efficiency (fault free state).
UNKNOWN_STATE	Is a certainty factor of unknown state of the process.
Isolated_Faults	Faults that have non-zero certainty factor are indicated in this element. For each fault its: name and ID (ID), Isolated and Present flags as well as current fault certainty factor (Current_Value), status and time stamp of its calculation are displayed.
Subprocess_FISes	Information about all SubFISes configured for current GlobalFIS are located in this element.
SUBPROCESS_FIS	Each SubFIS defined during configuration has one section describing its status. SubFISes are identified by unique ID. ID is displayed next to SubFIS name (in brackets). Status is a general description of SubFIS.
STATE_OF_COMPLETE_EFFICIENCY	Is a certainty factor of the state of complete efficiency (fault free state).
UNKNOWN_STATE	Is a certainty factor of unknown state of the process.
Isolated_Faults	Faults that have non-zero certainty factor are indicated in this element. For each fault its: name and ID (ID), Isolated flag as well as current fault certainty factor (Current_Value), status and time stamp of its calculation are displayed.
Thread_FISes	Information about all ThreadFISes (isolation threads) currently running in current SubFIS are located in this element.
THREAD_FIS	Each existing ThreadFIS has one section describing its status. ThreadFISes are identified by automatically generated unique ID. Initiation time indicates time stamp when the thread was created. Initiator is a indicator of diagnostic signal that triggered the thread. Its unique ID is displayed next to its name. Maximum isolation time is a time to achieve final diagnosis (if it is possible). Mode is a current isolation algorithm mode. Status is a general description of ThreadFIS.
Useful_Diagnostic_Signals	Diagnostic signals used in the current ThreadFIS are indicated in this element. For each fault its: name and ID (ID), 'Determined' and 'is_detection' flags and current value of 'cumulative symptom time' are displayed.
Possible_Faults	Faults used in the current ThreadFIS are indicated in this element. For each fault its: name and ID (ID) and Rejected flag are displayed.
Diagnosis	Faults that have non-zero certainty factor are indicated in this element. For each fault its: name and ID (ID), Isolated flag as well as current fault certainty factor (Current_Value), status and time stamp of its calculation are displayed.
Diagnostic_Rules	This element groups detailed information about diagnostic rules that are used in the current isolation thread.

4.2. Called external RPCs

General FDI Messages produced by ‘RTFAULTISOLATOR’ (see section 3) are published under ‘RPC’ topic as RPC function ‘*DisplayMessage*’. Syntax of the message is presented in Figure 8. Exemplary message is shown on Figure 8.

```
<Rpc>
  <Rpc_Name>DisplayMessage</Rpc_Name>
  <Rpc_Params>
    <Rpc_Param>
      <Value>
        <String>sub_fis_2</String>
      </Value>
    </Rpc_Param>
    <Rpc_Param>
      <Value>
        <String>thread_fis_2</String>
      </Value>
    </Rpc_Param>
    <Rpc_Param>
      <Value>
        <String>Creating new isolation thread</String>
      </Value>
    </Rpc_Param>
    <Rpc_Param>
      <Value>
        <String>2004-02-14T17:01:53:005</String>
      </Value>
    </Rpc_Param>
  </Rpc_Params>
</Rpc>
```

Figure 8. Structure of the “*DisplayMessage*” RPC

Table 4. The available FDI Messages content

RPC name	RPC Parameters		
	#	Type	Value / Description
<i>DisplayMessage</i>	1	String	subsystem FIS ID
	2	String	thread FIS ID
	3	String	message content

5. iFuzzyFDI – FIS Configurator GUI's

The iFUZZYFDI - FIS CONFIGURATOR has several GUI's used by the user to set up the module configuration parameters. The map of the GUI's is presented on Figure 9. Particular GUI's are described in the following sections.

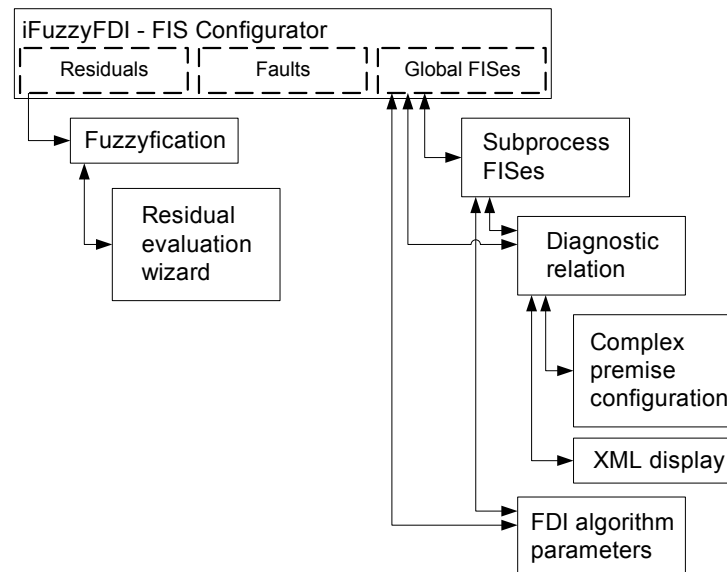


Figure 9. iFUZZYFDI - FIS CONFIGURATOR GUI's scheme

5.1. Residuals and Residual evaluation wizard

This screen is used to set up residuals parameters. There are two kinds of residuals:

- ♦ **Model based.** These residuals are managed (created, removed) by other AMANDD modules. Here, the user can only change residual parameters. They are indicated by checked 'model based' check box.
- ♦ **Non-model based.** These are all other residuals. They can be managed by other AMANDD modules as well as though FIS CONFIGURATOR. The user can add new residual, clone selected one or remove all selected (called from menu, toolbar or popup menu).

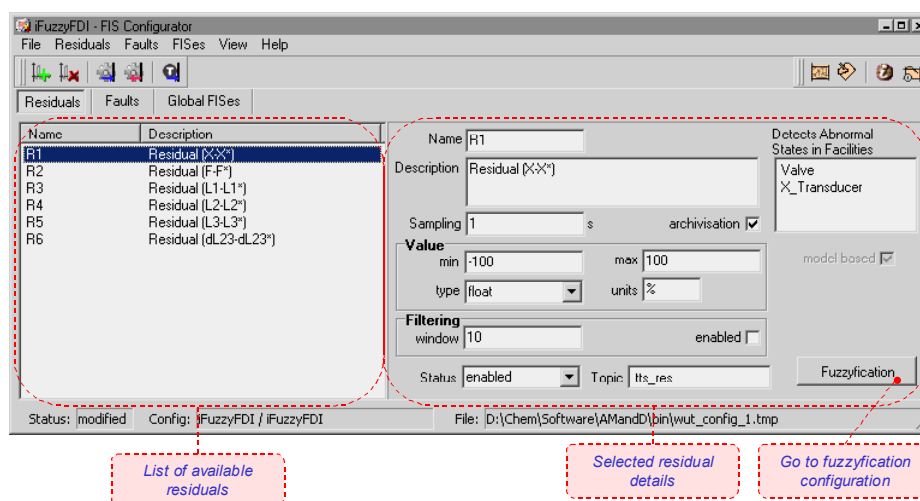


Figure 10. Residuals configuration GUI.

The list of all available in the system residuals is visible at the left side on the screen. The list of fields with residual configuration details is visible at the right side.

Table 5. Main residuals parameters.

Name	Values	Description
<i>Name</i>	valid XML attribute format	This attribute is used by the user to recognise residual, so it should be short but meaningful. However, from the system point of view, it does not have to be unique.
<i>Filtering</i>		
<i>window</i>	positive integer greater than 0	The width of averaging window (in samples, 1 means no averaging).
<i>enabled</i>	flag	Switches on / off filtering.
<i>Status</i>	<i>enabled</i> <i>disabled</i>	The disabled residuals are removed from calculations. Can be used for temporary switching off the residual, <i>e.g. due to bad residual behavior caused by modeling errors or changes in the process structure but not in respect to existing faults.</i>
<i>Topic</i>	one of the predefined in the system	The residual will be published under this topic. By setting different topics one can divide published residuals into separate messages, <i>e.g. groups of residuals corresponding to process sections.</i>

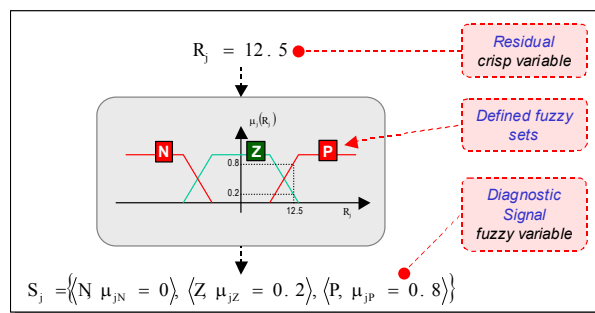
All the important residual parameters with description are shown in Table 5. The rest of the parameters have informative meaning and is not very important on the current stage of toolbox development.

Changes done in the fields during editing are stored when the **Enter** key is pressed or when the focused is moved to another field. Pressing the **Esc** will restore current field value – all the changes will be lost. To add new connection to the facility or to the topic one must press **Ins** or double-click the items list. To disconnect from facilities the user must select desired facilities on the list and press **Del**. All the commands can be called also form menu or toolbar.

By double-clicking on the residuals list or by ‘Fuzzyfication’ button one can switch to **Fuzzyfication GUI**.

5.2. Fuzzyfication

One of the most important things concerning residual configuration is their fuzzyfication method. The detection and isolation algorithms deals with diagnostic signals which are quantitative (fuzzy) representation of residuals.

**Figure 11. Diagnostic signals – qualitative (fuzzy) representation of residuals.**

The user must choose the number and parameters of the fuzzy sets. There is no automatic method for doing this, however one can base on statistical parameters of residuals signal in normal (fault free) state. The most important is proper positioning of fuzzy set describing normal residual values (in fault free state). Only one such a fuzzy set can be define, its kind have to set to ‘*positive*’. Exceeding the range of values covered by this fuzzy set testify about abnormal process state – fault detection is notified and isolation procedures are started.

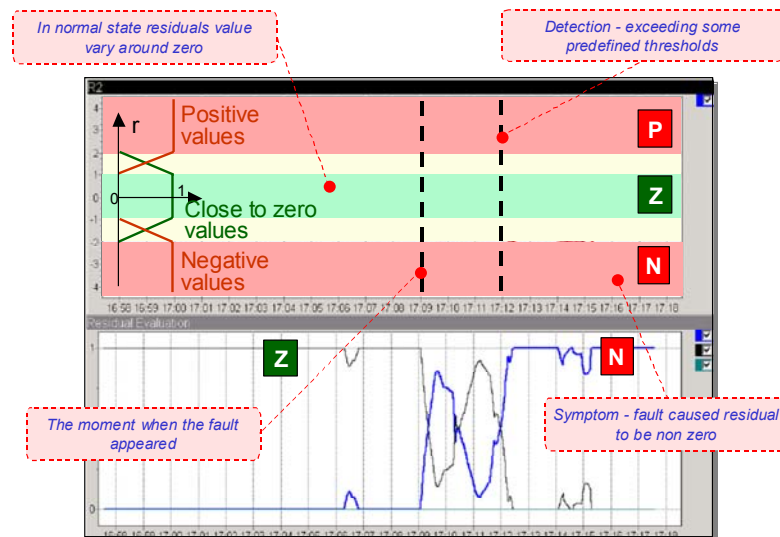


Figure 12. Example of residual evaluation.

The number of fuzzy sets describing abnormal residual values (in states with faults) depends on current application. Usually, one or two (for positive and negative values) additional fuzzy sets are applied.

All the important fuzzy sets parameters with description are shown in Table 6.

Table 6. Main fuzzy sets parameters.

Name	Values	Description
<i>Name</i>	valid XML attribute format	This attribute is used by the user to identify fuzzy sets during configuration of the diagnostic relation, so it should be short but meaningful, e.g. 'Z', 'P' and 'N' or '0' and '1'.
<i>Kind</i>	<i>positive</i>	Indicates the fuzzy set covering normal residual values (in fault free state). For one residual only one such a fuzzy set can exist.
	<i>negative</i>	Indicates the fuzzy sets representing abnormal residual values (in states with faults).
<i>MF Type</i>	<i>multi-line</i> , <i>trapezoid</i> , <i>slope</i>	Defines the type of membership function describing fuzzy set. Each type has different shape and different number of characteristic points that user must set up.

Residuals fuzzyfication parameters are set up through **Fuzzyfication** GUI. The name of currently configured residual is displayed in the window title. The window is divided into three sections:

- ◆ **Normal residual values description.** The list of statistical parameters of the residual and its histogram are displayed in the left-bottom corner of the screen, if they are available. They can be used during setting membership functions (MF) parameters.
- ◆ **Fuzzy sets configuration.** The list of defined fuzzy sets and their parameters are in bottom-right part of the screen. The appearance of the 'MF Parameters' section depends on the selected type of MF. The meaning of particular MF parameters is explained on Figure 13, Figure 14 and Figure 15.
- ◆ **Membership function plot.** Graphical representation of defined fuzzy sets (membership function plot) is placed in the top part of the screen. For each fuzzy set separate line is drawn. One of the fuzzy sets that is active is drawn with bold line. The characteristic points of active membership function are marked with big dot. This dots are used to manipulate the parameters of active fuzzy set membership function.

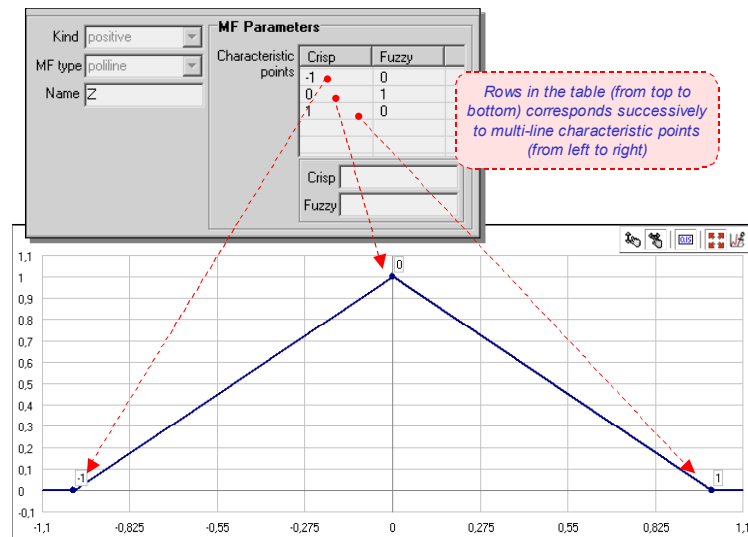


Figure 13. Meaning of the multi-line MF parameters.

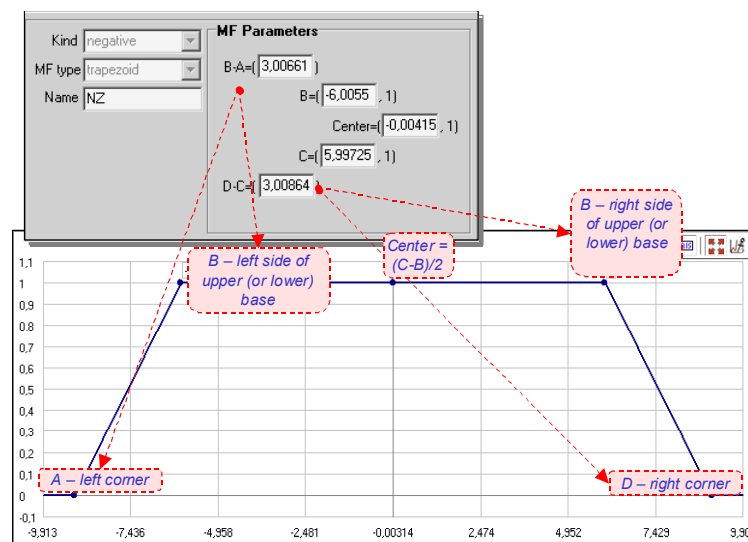


Figure 14. Meaning of the trapezoid MF parameters.

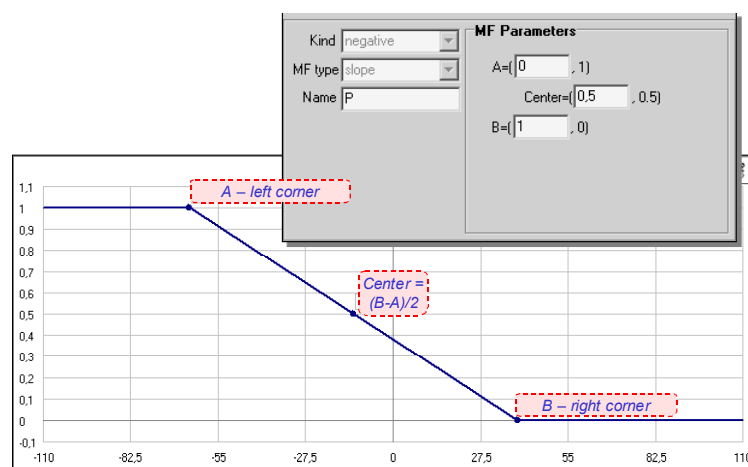


Figure 15. Meaning of the slope MF parameters.

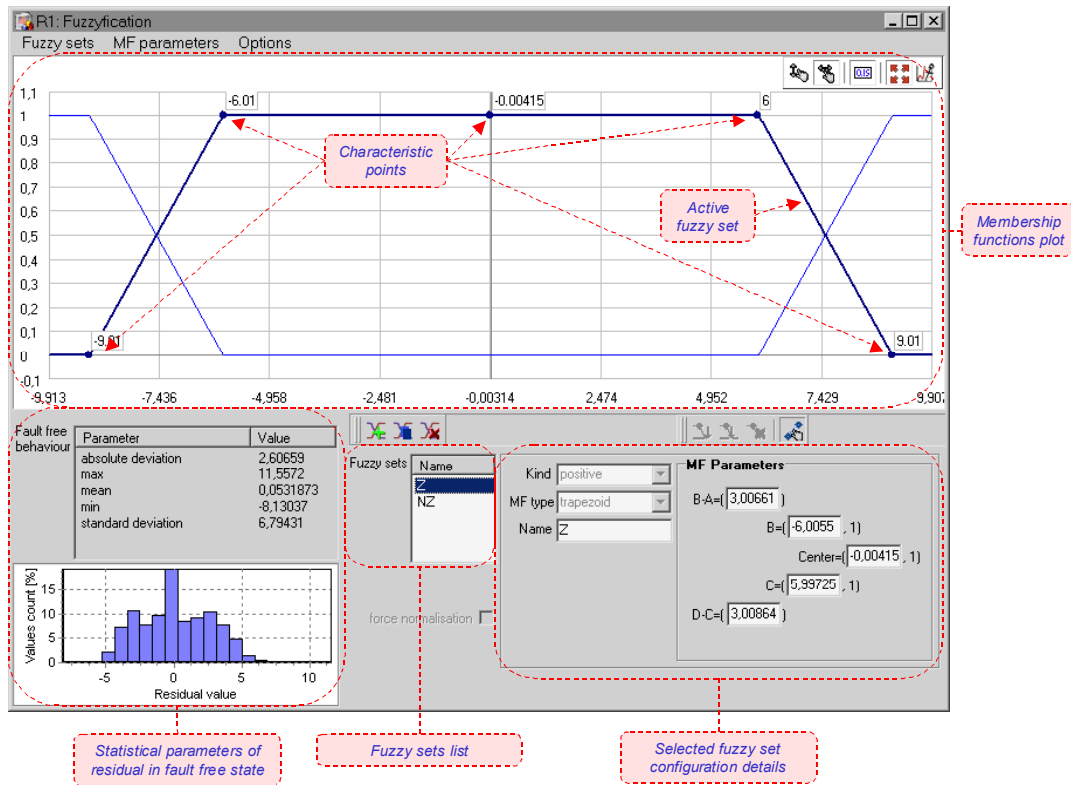


Figure 16. Fuzzyfication GUI

The user can change membership function parameters by clicking and dragging selected characteristic point. The number and position of characteristic point depends on the kind of membership function:

- ◆ 'multi-line'. Each point of the multi-line has one corresponding characteristic point displayed on the plot. During dragging the point the user directly changes the 'crisp' (x) and 'fuzzy' (y) parameters of multi-line point.
- ◆ 'trapezoid'. This membership function has five characteristic points: one per each corner plus additional one placed in the centre of upper base of the trapezoid. During dragging the point the user changes values of several membership function parameters.

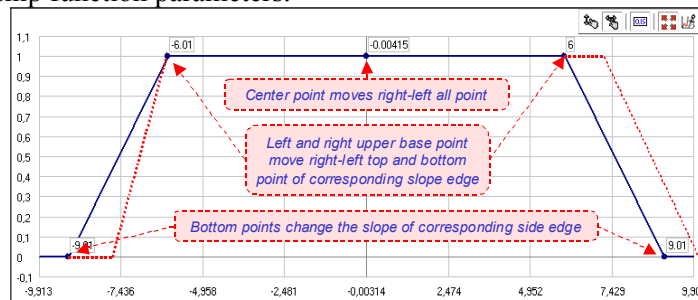


Figure 17. Changing trapezoid MF parameters through characteristic point dragging

- ◆ 'Slope'. This shape has three characteristic points: one per each corner plus additional one placed in the centre of sloping edge. During dragging the point the user changes values of several membership function parameters.

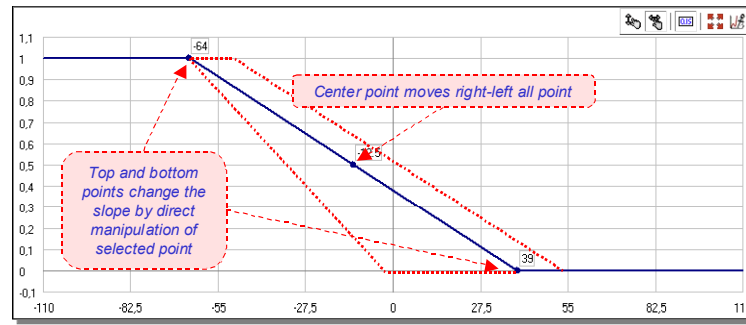


Figure 18. Changing slope MF parameters through characteristic point dragging

The user can add new fuzzy set, clone selected one or remove all selected (called from menu, toolbar or popup menu) under following conditions:

- ◆ Only one fuzzy set of 'positive' kind can exist.
- ◆ The fuzzy set of 'positive' kind can not be removed if considered residual belongs to any of the **Global FISes** (is used in diagnostic relation).
- ◆ If considered residual belongs to any of the **Global FISes** (is used in diagnostic relation) than removing the fuzzy set of 'negative' kind will cause removing corresponding links to that fuzzy set defined in the diagnostic relations.

During manipulation of characteristic points the following options (available through menu, plot popup menu or toolbar) are useful:

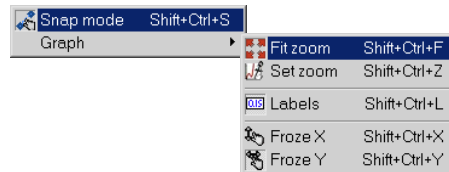


Figure 19. Options of the membership functions plot and manipulations.

- ◆ **Snap mode.** If this option is switch on than changes made to one of the fuzzy sets automatically influence adjacent fuzzy set. The membership functions of the following and the next fuzzy sets are modified in that way to keep the sum of all membership functions for each crisp (x) point equal 1.
- ◆ **Fit zoom.** The X zoom of the plot is automatically kept to the range between minimal and maximal crisp values of all characteristic points plus extra 10%.
- ◆ **Set zoom.** This command invokes dialog box for setting desired zoom range.
- ◆ **Labels.** If the option is switch on the characteristic points labels are visible.
- ◆ **Froze X.** If the option is switch on only fuzzy (y) values of dragged characteristic point are changing.
- ◆ **Froze Y.** If the option is switch on only crisp (x) values of dragged characteristic point are changing.

The user can define fuzzy sets manually one-by-one or automatically with use of **Fuzzyfication Wizard**. Automatic creation is called by 'Create evaluation set' menu item. Wizard can be started only there are no fuzzy sets defined yet. The screens of fuzzyfication wizard are shown on Figure 20.

- ◆ Firstly, the user defines how the fuzzy set of 'positive' kind (describing normal residual values) will be created. Its width can be calculated based of standard deviation, minimal-maximal values or entered manually. The width of fuzzyfication range between fuzzy sets is defined by the 'fuzzifness' parameter. The meanings of the parameters are shown on Figure 21.

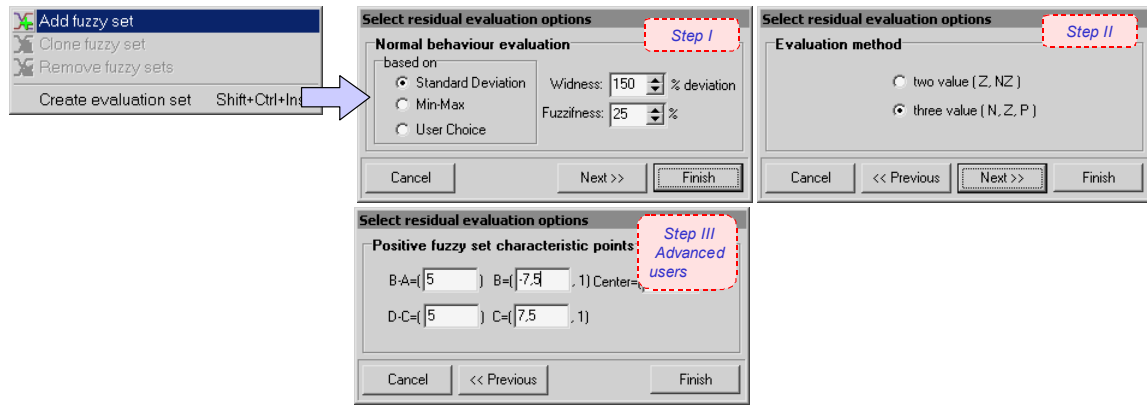


Figure 20. Fuzzyfication wizard.

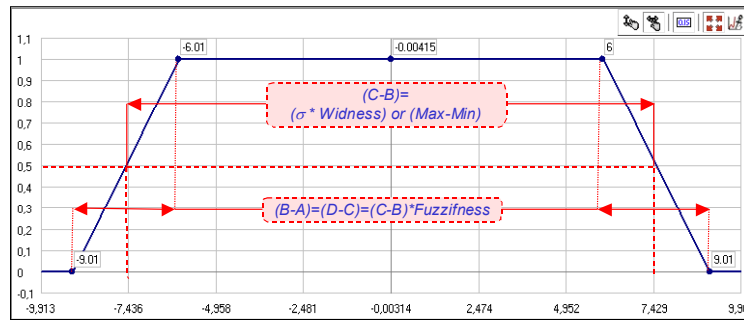


Figure 21. Setting fuzzy set trapezoid MF parameters based on standard deviation and min-max residual values.

- ◆ Then, the user choose if two- or three-value evaluation should be done. Default names for created fuzzy sets will be attributed.
- ◆ Third page is for advanced users. It allows to manually adjust positive fuzzy set MF parameters.

5.3. Faults

This screen is used to set up faults parameters. The user can add new fault, clone selected one or remove all selected (called from menu, toolbar or popup menu).

The list of all considered faults in the system is visible at the left side on the screen. The list of fields with fault configuration details is visible at the right side.

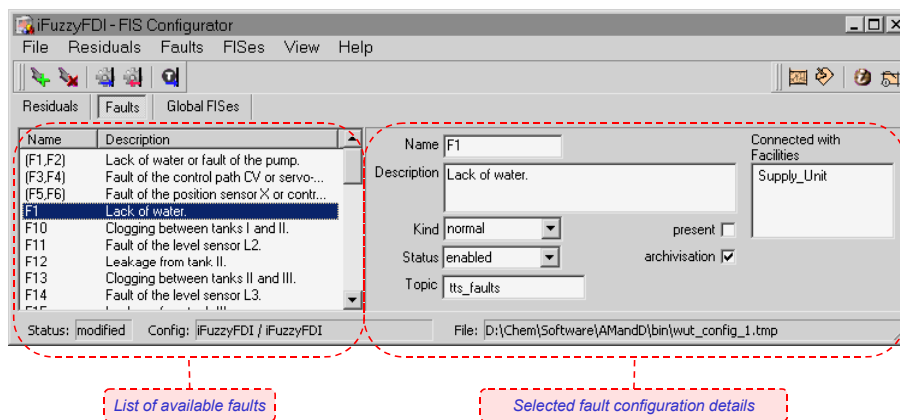


Figure 22. Faults configuration GUI.

All the important fault parameters with description are shown in Table 7. The rest of the parameters have informative meaning and is not very important on the current stage of toolbox development.

Table 7. Main faults parameters.

Name	Values	Description
<i>Name</i>	valid XML attribute format	This attribute is used by the user to recognise fault, so it should be short but meaningful. However, from the system point of view, it does not have to be unique.
<i>Kind</i>	<i>normal</i>	All specific faults defined by the expert has this kind
	<i>ok</i>	Special fault that in fact indicates the certainty factor of the state of complete efficiency . Automatically generated rule of such a state will be attributed to the fault of 'ok' kind, if such one exists. Only one fault of that kind can exists in each GlobalFIS.
	<i>us</i>	Special fault that indicates the certainty factor of the unknown state of the process . Such fault is pointed out in the diagnosis if observed symptoms pattern does not match those ones defined by the expert in diagnostic relation. Certainty factor of unknown state will be automatically attributed to the fault of 'us' kind, if such one exists. Only one fault of that kind can exists in each GlobalFIS.
<i>Status</i>	<i>enabled</i> <i>disabled</i>	The disabled faults are removed from calculations. Can be used for temporary switching off the fault.
<i>Topic</i>	one of the predefined in the system	The fault will be published under this topic. By setting different topics one can divide published faults into separate messages, <i>e.g. groups of faults corresponding to process sections</i> .
<i>Present</i>	flag	Used to marked fault as present. Such fault is excluded from reasoning however is pointed out in the diagnosis with certainty factor equal 1 (independently from observer symptoms). Mainly used for manual reconfiguration for longer periods of time.

Changes done in the fields during editing are stored when the **Enter** key is pressed or when the focused is moved to another field. Pressing the **Esc** will restore current field value – all the changes will be lost. To add new connection to the facility or to the topic one must press **Ins** or double-click the items list. To disconnect from facilities the user must select desired facilities on the list and press **Del**. All the commands can be called also form menu or toolbar.

5.4. Global FISes, Subprocess FISes and FDI Algorithm Parameters

The user must define one or more Global FISes. GlobalFIS joins the subset of residuals and the subset of faults that belong to some part of the process, *e.g. the division – evaporation station*. In many cases it is enough to declare one Global FIS. The FDI procedures are calculated completely independent for defined Global FISes. They must have disjoint subsets of residuals and faults.

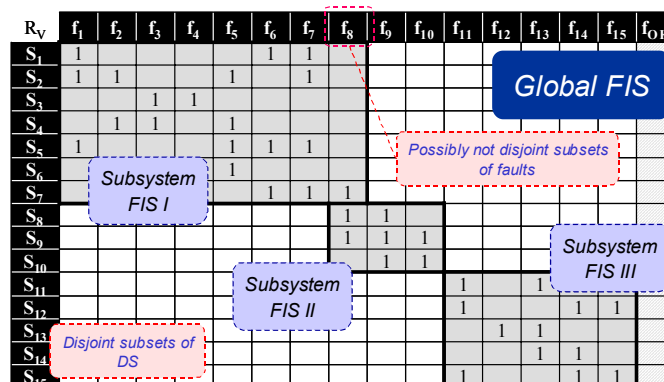


Figure 23. Division of Global FIS into Subprocess FISes.

All the important Global FIS parameters with description are shown in Table 7. The rest of the parameters have informative meaning and is not very important on the current stage of toolbox development. Additionally, the user must define diagnostic relation and can change FDI algorithm parameters. This procedures are described in other sections.

Table 8. Main GlobalFIS parameters.

Name	Values	Description
<i>Name</i>	valid XML attribute format	This attribute is used by the user to recognise GlobalFIS, so it should be short but meaningful. However, from the system point of view, it does not have to be unique.
<i>Residuals</i>	chosen residual ID's from available in the system	Each residual can belong to at most one Global FIS.
<i>Faults</i>	chosen faults ID's from available in the system	Each fault can belong to at most one Global FIS. Each Global FIS must have its own 'ok' and 'us' faults if the coefficients of the state of complete efficiency and the unknown state of the process are to be calculated.

The division of process into Global FISes is a top level decomposition. The next level is the decomposition into Subprocess FISes, *e.g. the evaporation station can be decompose into sections*. The user must decide if he want to decompose the Global FIS and how to do it.

For each Subprocess FIS the subset of residuals and faults from those defined for Global FIS must be chosen. All the important SubprocessFIS parameters with description are shown in Table 9. Additionally, the user can change FDI algorithm parameters. This procedure is described in other sections.

Table 9. Main SubprocessFIS parameters.

Name	Values	Description
<i>Name</i>	valid XML attribute format	This attribute is used by the user to recognise SubprocessFIS, so it should be short but meaningful. However, from the system point of view, it does not have to be unique.
<i>Residuals</i>	chosen residual ID's from those ones which were attributed to Global FIS	Each residual can belong to at most one Subprocess FIS.
<i>Faults</i>	chosen faults ID's from those ones which were attributed to Global FIS	Each fault can belong to several Subprocess FISes. Faults of 'ok' and 'us' kind are not attributed to any of the Subprocess FISes.

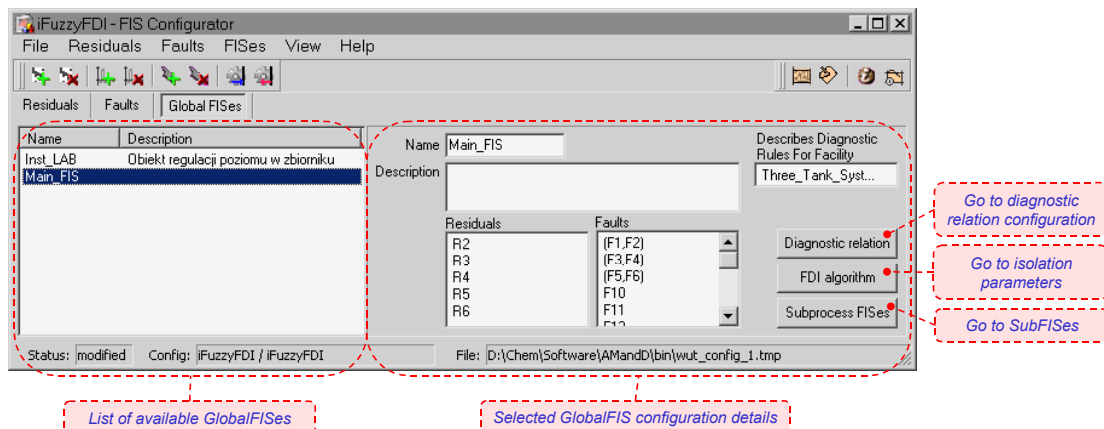


Figure 24. Global FISes configuration GUI.

The parameters of Global FISes are set-up through **Global FISes** GUI. The user can add new Global FIS, or remove all selected (called from menu, toolbar or popup menu). The list of all defined Global FISes is visible at the left side on the screen. The list of fields with Global FIS configuration details is visible at the right side.

Changes done in the fields during editing are stored when the **Enter** key is pressed or when the focused is moved to another field. Pressing the **Esc** will restore current field value – all the changes will be lost. To add new residual or fault one must press **Ins** or double-click the items list. To remove attributed items the user must select them and press **Del**. All the commands can be called also from menu or toolbar.

By double-clicking on the residuals list or by ‘Subprocess FISes’ button one can switch to **Subprocess FISes Configuration** GUI, by clicking ‘Diagnostic relation’ button to **Diagnostic relation** GUI. Clicking on ‘FDI algorithm’ button invokes dialogs for setting FDI algorithm parameters on Global FIS level. For more details see Section 7.

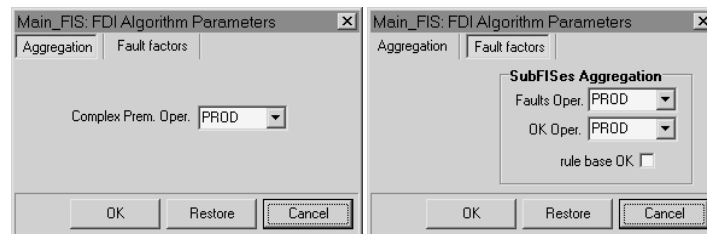


Figure 25. Dialogs for setting FDI algorithm parameters on the Global FIS level.

The parameters of Subprocess FISes are set-up through **Subprocess FISes** GUI. The user can add new Subprocess FIS, or remove all selected (called from menu, toolbar or popup menu). The list of all defined Subprocess FISes for particular Global FIS is visible at the left side on the screen. The list of fields with Subprocess FIS configuration details is visible at the right side.

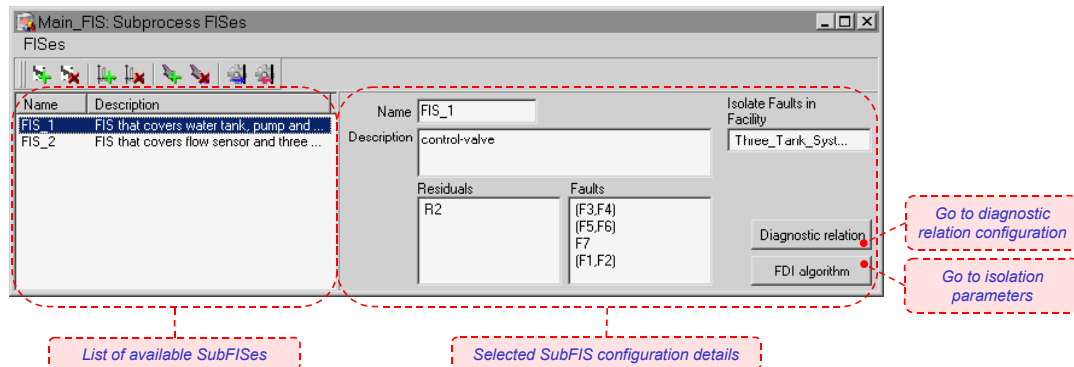


Figure 26. Subprocess FISes configuration GUI.

Changes done in the fields during editing are stored when the **Enter** key is pressed or when the focused is moved to another field. Pressing the **Esc** will restore current field value – all the changes will be lost. To add new residual or fault one must press **Ins** or double-click the items list. To remove attributed items the user must select them and press **Del**. All the commands can be called also from menu or toolbar.

By clicking ‘Diagnostic relation’ button one can switch to **Diagnostic relation** GUI. Clicking on ‘FDI algorithm’ button invokes dialogs for setting FDI algorithm parameters on SubFIS level. For more details see Section 7.

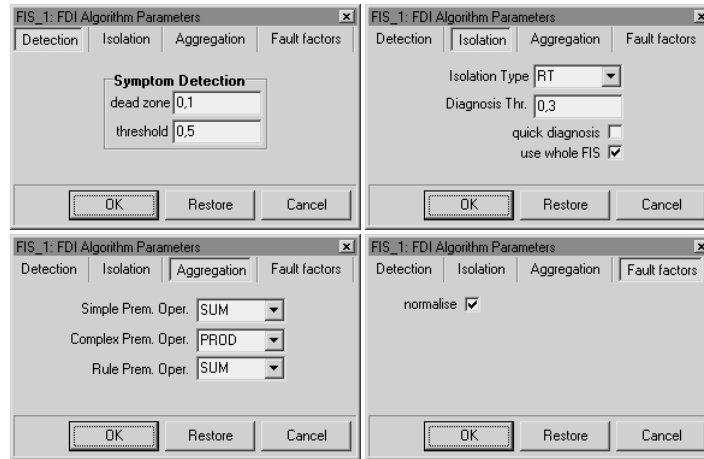


Figure 27. Dialogs for setting FDI algorithm parameters on the SubFIS level.

5.5. Diagnostic Relation and Complex Premise Configuration

The reasoning about system state (calculating fault certainty factors) is based on diagnostic rules that define the relation between observer symptoms (values of the residuals) and fault existing in the system. The structure of reasoning rule and used terminology are explained with the exemplary rule shown on Figure 28. The set of diagnostic rules is called diagnostic relation. It can be described by information system FIS. Diagnostic relation (FIS system) can be written in a form of a table. Each diagnostic rule corresponds to one of the table columns.

Configuring diagnostic relation (faults-symptoms-time) relation is the most important task to be done by the user during toolbox configuration. This can be done through **Diagnostic Relation** GUI. The user must specify the relation between faults and observed symptoms based on expert knowledge taking into account such elements like:

- ♦ the structure and type of models that are used for residual generation, in general the residual equation structure,
- ♦ the knowledge about the process and its sub-elements properties,
- ♦ careful analysis of particular cases.

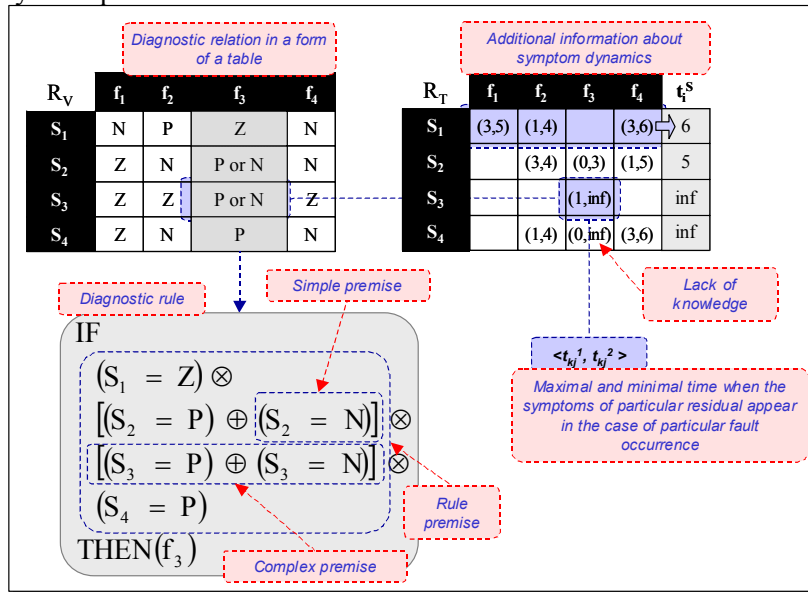


Figure 28. The structure of diagnostic rules, used terminology. FIS describing diagnostic relation that can be shown in a form of a table.

Additionally time limits for symptoms appearing can be defined for each pair <diagnostic signal, fault>. They specify the minimal and the maximal time when the symptoms of particular diagnostic signal (residual) can be observed when the particular fault appeared in the system. If no such information is available the user can set:

- ◆ ‘0’ for minimal symptoms time, means that the symptoms can appear even immediately when the fault emerged,
- ◆ ‘inf’ for maximal symptoms time.

The diagnostic rules can be formulated in a very flexible way: each complex premise can have more than one fuzzy set attributed, several rules for the same fault can be defined.

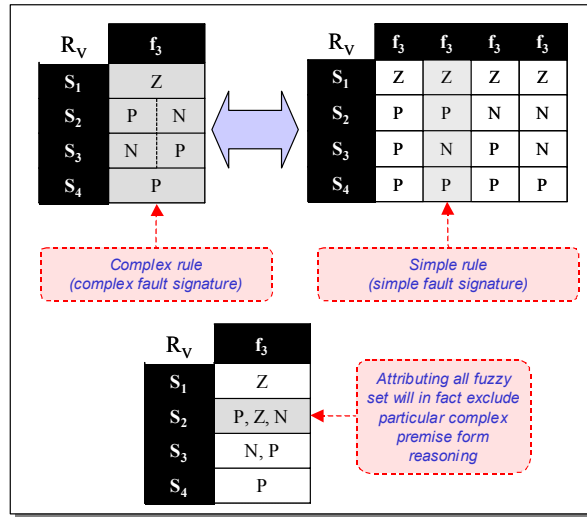


Figure 29. The user can form diagnostic rules in a very flexible way.

The diagnostic relation is defined on the Global FIS level through **Diagnostic Relation** GUI. The name of current Global FIS is displayed in the window title.

The interface is divided into two sections: diagnostic relation display in the form of a matrix and FDI report. It can work in two modes:

- ◆ **Editing mode.** The user can edit diagnostic relation. This mode is active when interface is launched from Global FISes GUI.
- ◆ **Analysis mode.** The user can not edit diagnostic relation. This mode is active when interface is launched from Subprocess FISes GUI.

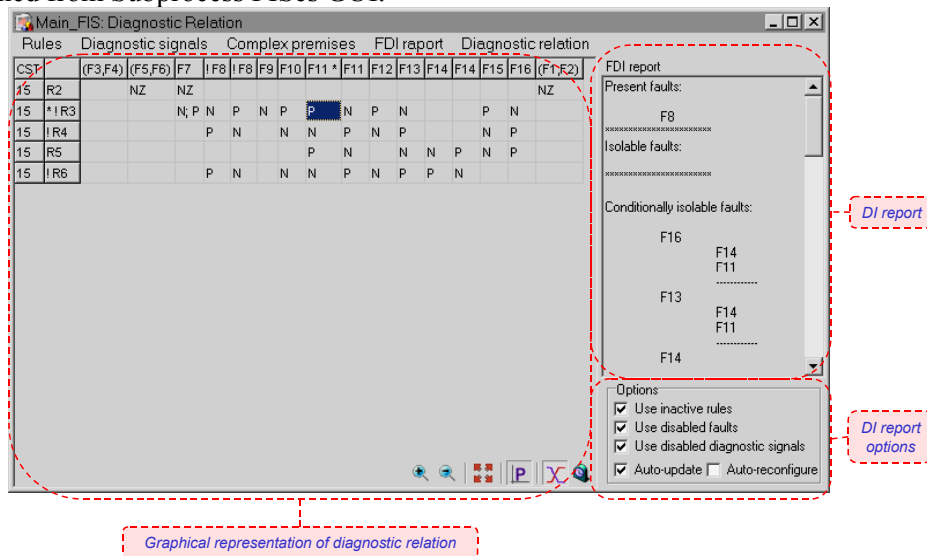


Figure 30. Diagnostic relation GUI.

Displayed table represent the diagnostic relation in a form of a matrix. Each column of a table corresponds to one of the diagnostic rules. The column headers represent the rule. They display the name of the fault represented in rule consequence. Particular table cells represent complex premises (for each pair <residual, fault>). Row headers represent the residuals (diagnostic signals) – all complex premises in the row concern that diagnostic signal. There is one special column named 'CST'. This column shows *cumulative symptom times* calculated for particular residuals.

	CST	* F7	! F8	! F8	F11	* F9
20 R1						
20 R2		NZ				
40 * ! R3		N; P	N	P	P	N
25 ! R4			P	N	N	

Callouts in Figure 31:

- Disabled fault:** * F7
- Present fault:** ! F8
- Inactive rule:** ! F8
- Suspended residual:** 20 R1
- Disabled residual:** 40 * ! R3
- Calculated residual cumulative symptom times:** 25
- Residuals designation:** R1, R2, R3, R4
- Diagnostic rule:** F7, F8, F11, F9
- Attributed complex premise symptoms:** NZ, N, P, N
- Notification of faults in rules consequence:** * F9
- Active complex premise:** N (in row R3, column F9)

Figure 31. Diagnostic relation matrix: structure and marking.

There are several special marking next to residuals and fault names:

- * – indicates disabled residuals, faults and inactive rules,
- ! – indicates present faults and suspended residuals.

The user can modify diagnostic relation by double-clicking (or selecting proper menu item) complex premise (one of the table cells) to be modified. This will invoke *complex premise configuration* dialog shown on Figure 32.

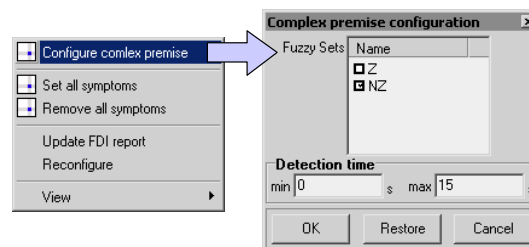


Figure 32. Complex premise configuration dialog.

With use of this dialog one can select simple premises that should be included in considered complex premise. One can also set detection constraints.

During manipulation of diagnostic relation the following options (available through menu, plot popup menu or toolbar) are useful:

- Only symptoms mode. In this mode the complex premises (cells) where no symptoms are attributed are displayed as blank ones.
- Symptoms mode. In this mode simple premises are displayed.
- Time constraints mode. In this mode time constraints of complex premises are displayed.
- Zoom in. Increase zoom by predefined factor (10%).
- Zoom out. Decrease zoom by predefined factor (10%) unless minimal zoom is achieved.
- Fit zoom. Find maximal zoom to fit the whole table in current window size.

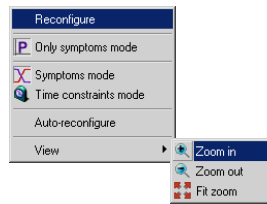


Figure 33. Options and controls of the diagnostic relation display.

There are several procedures available that can speed up diagnostic relation configuration. They are available through menu and popup menus. Different popup menus are invoked depending on the mouse position over the diagnostic matrix.

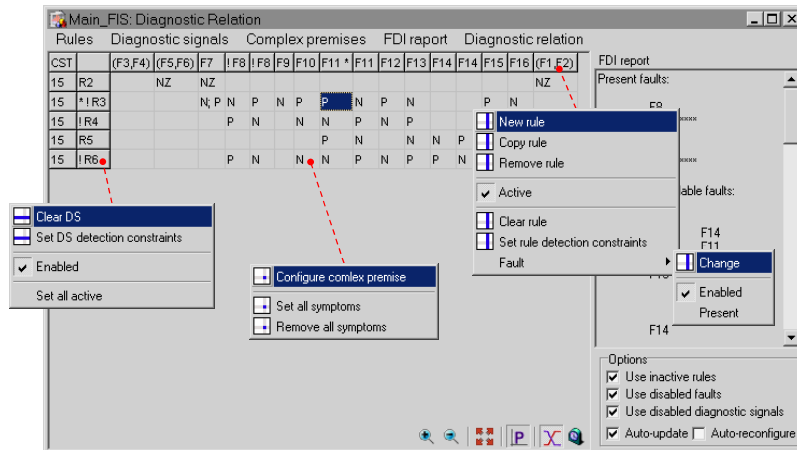


Figure 34. Different popup menus for diagnostic relation matrix.

The popup menu for column headers includes procedures for fault and rules manipulation:

- ◆ **New rule.** Creates new rule for the fault specified by the user.
- ◆ **Copy rule.** Creates a copy of a rule. Rule fault and all complex premises in the new rule are the same as in original one.
- ◆ **Remove rule.** Removes current rule from diagnostic relation.
- ◆ **Active.** Display and switch the rule 'active' attribute. Inactive rules are excluded from on-line calculations.
- ◆ **Clear rule.** Remove all the symptoms from all complex premises of current rule. After this operation a corresponding positive fuzzy sets are attributed to complex premises of current rule.
- ◆ **Set rule detection constraints.** Change the detection constraints for all the complex premises of current rule that have attributed at least one symptom. Proper dialog for setting this constraints is displayed.
- ◆ **Fault / Change.** Change the fault attributed to current rule consequence. Proper dialog for selecting new fault is displayed.
- ◆ **Fault / Enabled.** Display and switch the status of a fault attributed to the rule consequence. Disabled faults (and the rules attributed to them) are excluded from on-line calculations.
- ◆ **Fault / Present.** Display and switch the 'present' flag of a fault attributed to the rule consequence. Present faults (and the rules attributed to them) are excluded from on-line calculations – such faults have automatically attributed certainty factor equal 1.

Existing (present) faults in the system cause some residuals to be temporary useless for father process monitoring. The user can call reconfiguration procedure (from menu or popup menu) to determine which residuals will be suspended. If the 'auto-reconfiguration' option is switch on than the reconfiguration is done automatically.

The popup menu for row headers includes procedures for manipulations of diagnostic signal and corresponding complex premises of all rules:

- ◆ **Clear DS.** Remove all the symptoms from complex premises of current residual from all rules (whole row). After this operation a corresponding positive fuzzy sets are attributed to these complex premises.
- ◆ **Set DS detection constraints.** Change the detection constraints for all the complex premises of current residual from all rules (whole row) if at least one symptom is attributed. Proper dialog for setting this constraints is displayed.
- ◆ **Enabled.** Display and switch the status of a residual. Complex premises which residual is disabled are excluded from on-line calculations.
- ◆ **Set all active.** Switch all residuals mode from 'suspended' to 'active'.

The popup menu for cells includes procedures for complex premise manipulation:

- ◆ **Configure complex premise.** Opens *complex premise configuration* dialog.
- ◆ **Set all symptoms.** Attributes to complex premise all symptoms of its residual.
- ◆ **Remove all symptoms.** Removes from current complex premise all symptoms attributed to it. After this operation only a positive fuzzy sets is attributed that complex premises.

The FDI report includes results of the analysis considering fault detectability and isolability. It includes information about present, isolable, conditionally isolable, unisolable, undetectable faults and suspended diagnostic signals. FDI report updating procedure can be called from menu item or can be done automatically. There are several options that control the FDI report:

- ◆ **Use inactive rules.** If the option is switch off than the inactive rules are not taken into account during FDI report update.
- ◆ **Use disable faults.** If the option is switch off than the disabled faults and the rules attributed to them are not taken into account during FDI report update. Such faults are automatically assigned to the group of undetectable faults.
- ◆ **Use disabled diagnostic signals.** If the option is switch off than the complex premises of disabled diagnostic signal are not taken into account during FDI report update.
- ◆ **Auto-update.** If the option is switch on then the FDI report is updated automatically.

6. iFuzzyFDI – RT Fault Isolator GUI's

The iFUZZYFDI – RT FAULT ISOLATOR has several GUI's used by the user to set up the module configuration parameters. The map of the GUI's is presented on Figure 9. Particular GUI's are described in the following sections.

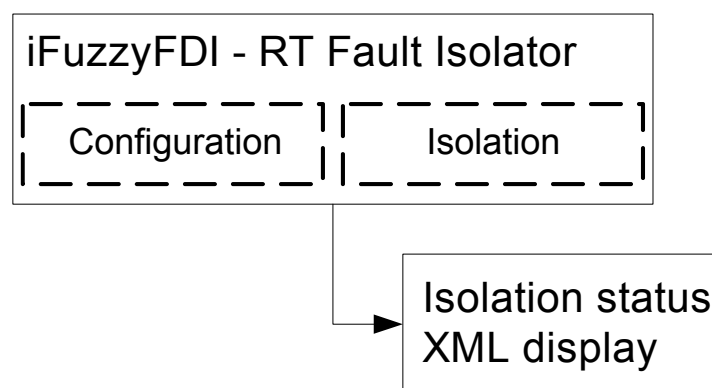


Figure 35. iFUZZYFDI – RT FAULT ISOLATOR GUI's scheme

GUI of on-line calculation module RT FAULT ISOLATOR is mainly used for monitoring the status of the FDI algorithm. However, the user can also change some of the configuration parameters (all this features are available through RPCs).

6.1. Isolation

Monitoring of FDI algorithm status can be done through 'Isolation' page.

The faults with greater than 0 certainty factor are displayed in the upper list of that page. They are displayed: fault name, value (current certainty factor), status and origin. The origin is a set of ThreadFISes IDs which pointed out that fault.

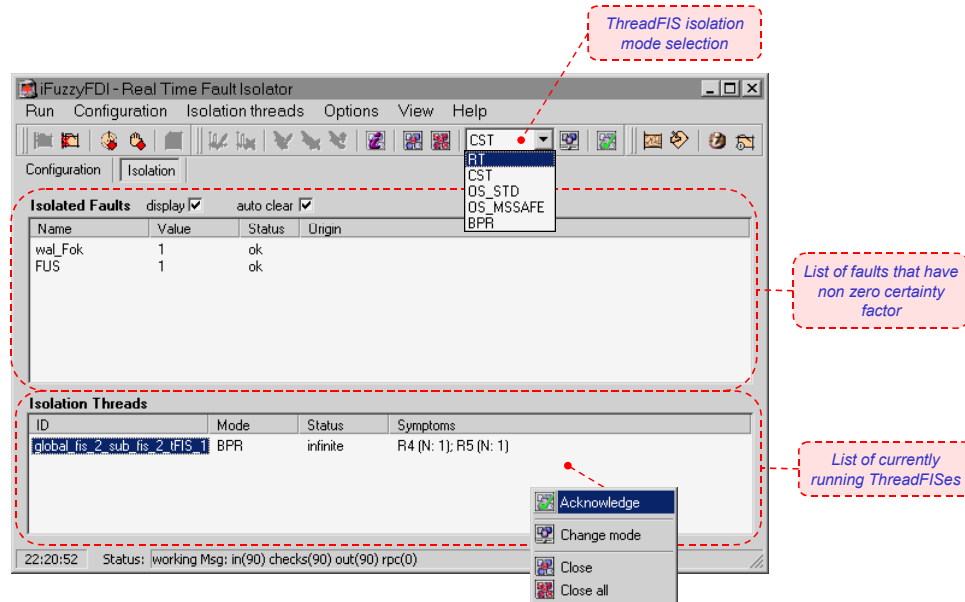


Figure 36. Isolation page of RTFAULTISOLATOR GUI.

The lower list displays currently running isolation threads (*ThreadFISes*). The list contains thread: ID, current isolation mode, status and the symptoms list. Symptoms list includes all the observer symptoms (with greater than 0 activation level) of the residuals that are used in corresponding isolation thread.

The user can control isolation threads by calling proper procedures (from menu or popup menu):



Acknowledge. When the user acknowledge the isolation thread than all the existing faults pointed out by this thread are marked as *isolated*. They remain in this state until the fault decay will be detected (when all observed symptoms fade).



Change mode. This procedure switches selected isolation thread mode to the one chosen in the mode selection combo box (visible on the toolbar).



Close. This procedure sends command to close selected isolation thread.



Close all. This procedure sends command to close all existing isolation threads.

6.2. Configuration

Changes to on-line configuration can be done through 'Configuration' page.

All residuals used in at least one GlobalFIS are displayed in the list on left side. They are displayed residual name, status and suspended property. The user can send command (from menu, toolbar or popup menu) to disable  or enable  selected residual. Disabled residuals are temporary excluded from calculations.

All faults considered in at least one GlobalFIS are displayed in the list on right side. They are displayed fault name, status and present and isolated properties. The user can send command (from menu, toolbar or popup menu) to disable  or enable  selected faults.

Disabled faults are not taken into account during calculations. One has also possibility to change 'present' property of the fault . Faults that are present are extruded from calculation and their certainty factor is automatically set to 1.

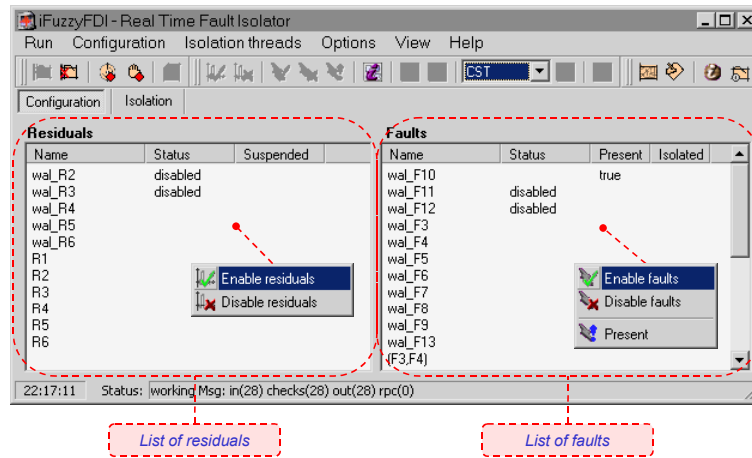


Figure 37. Configuration page of RTFAULTISOLATOR GUI.

6.3. Isolation Status XML Display

One has also possibility to see complete status of the isolation algorithm. To do this one must call 'Display status' procedure . Detailed description of the status information can be found in the section 4.1.

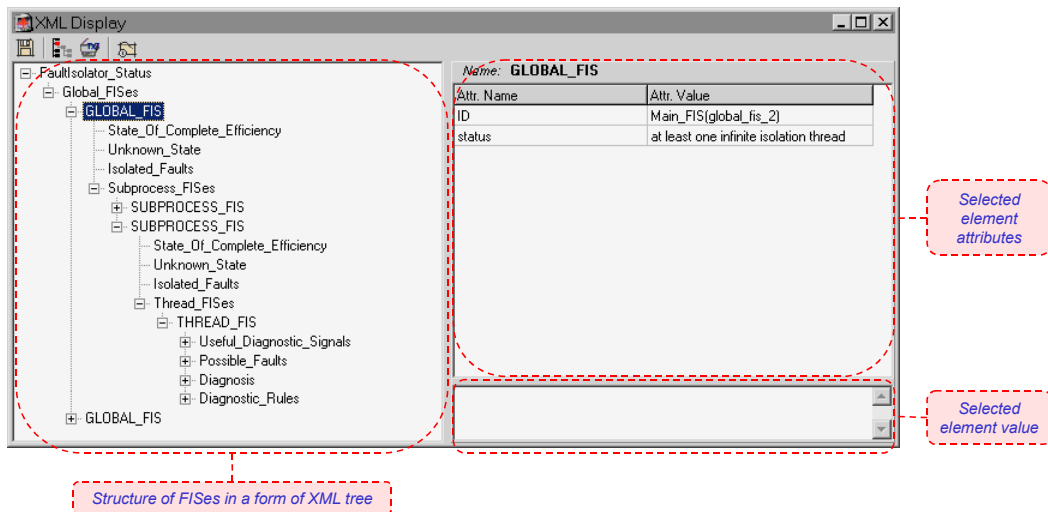


Figure 38. Dialog displaying isolation algorithm detailed status description.

7. FDI algorithm parameters

There are several parameters that control the execution of FDI algorithm. These parameters have predefined default values which should ensure proper algorithm behaviour. However, the user has the possibility to change these parameters. They are divided into three groups connected with detection, isolation, aggregation and calculating fault factors.

Table 10. FDI algorithm parameters of Detection category.

Name	FIS level	Values	Description
<i>Symptom detection</i>	Subprocess		Defines the level of membership function activation that triggers detection
<i>dead zone</i>		$<0, 0.5>$	One can specify dead zone to prevent against detection flashing, see Figure 39.
<i>threshold</i>		$<0, 1>$	The fault is detected when any of the symptoms activation levels is higher than $(threshold+dead_zone)$. The symptoms decay is detected when its activation level is lower than $(threshold-dead_zone)$. See Figure 39.

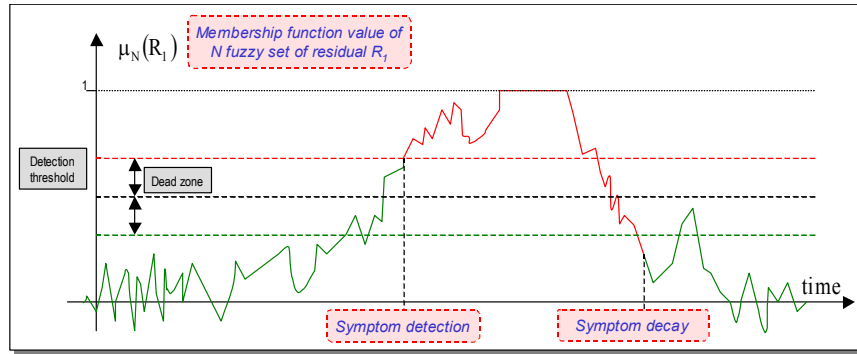


Figure 39. Dead zone in fault detection.

Table 11. FDI algorithm parameters of Isolation category.

Name	FIS level	Values	Description
<i>Isolation type</i>	Subprocess	<i>BPR</i>	The information about the symptoms dynamics (complex premises time constraints) is not taken into account.
		<i>OS_STD</i>	The information about the symptoms dynamics (complex premises time constraints) is not taken into account. Only diagnostic signals for which the symptoms are observed are considered. See Figure 40.
		<i>OS_MFSAFE</i>	The information about the symptoms dynamics (complex premises time constraints) is not taken into account. Only diagnostic signals for which the symptoms are observed are considered. Multi fault capability is switch on. See Figure 41.
		<i>CST</i>	Simplified information about the symptoms dynamics (complex premises time constraints) is taken into account – only cumulative symptoms times are calculated and used. See Figure 42.
		<i>RT</i>	Full information about the symptoms dynamics (complex premises time constraints) is taken into account. See Figure 43.
<i>Diagnosis threshold</i>	Subprocess	$<0, 1>$	The system automatically assumes that the fault exist in the system if its certainty factor exceeds diagnosis threshold. If there is a possibility the final diagnosis is elaborated.
<i>quick diagnosis</i>	Subprocess	boolean	Elaborates diagnosis as quick as possible even if time limits are not reached yet.
<i>use whole FIS</i>	Subprocess	boolean	Force to use all available diagnostic signals and considered faults during isolation thread creation. See Figure 44.

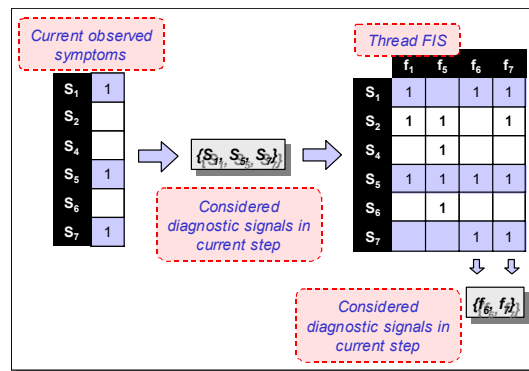


Figure 40. OS_STD isolation mode – taking into account only this diagnostic signals for which the symptoms are observed.

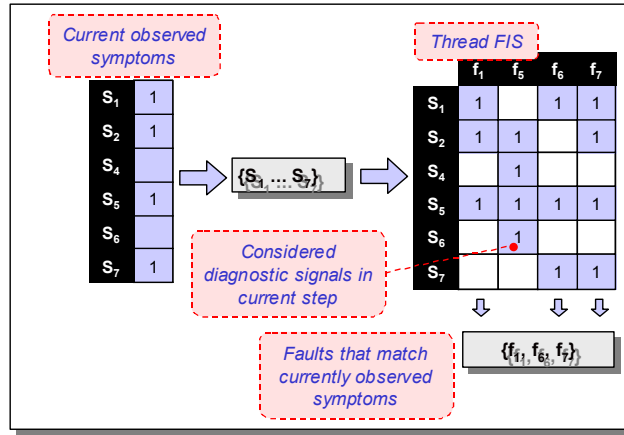


Figure 41. OS_MSAFE isolation mode – for each fault only those diagnostic signals are taken into account which symptoms are defined in the relation with that fault.

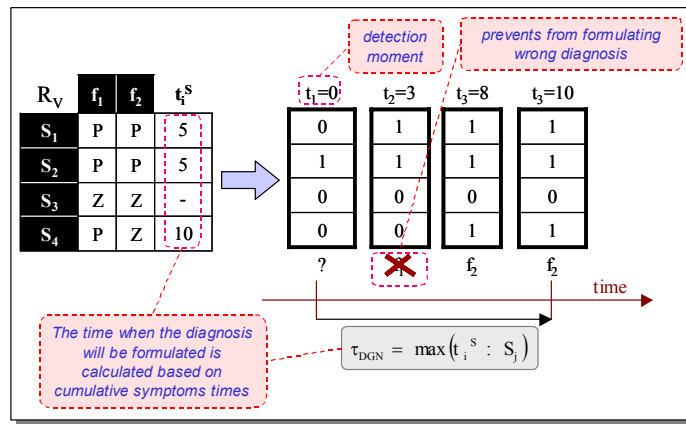


Figure 42. CST isolation mode – taking into account cumulative symptom times. It prevents from temporary incorrect diagnosis.

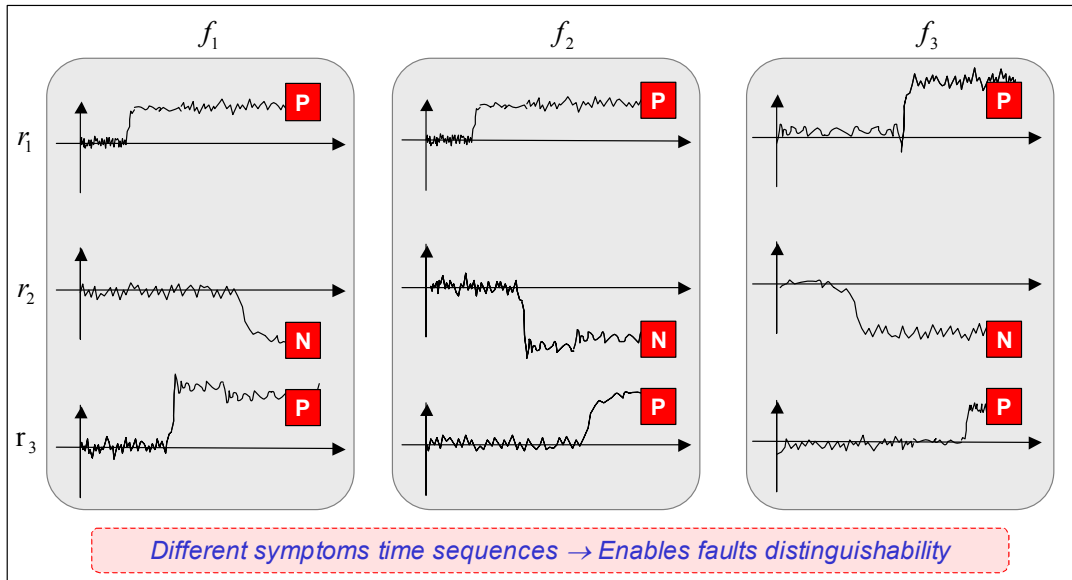


Figure 43. Taking into account symptoms dynamics can lead to increase the fault isolability.

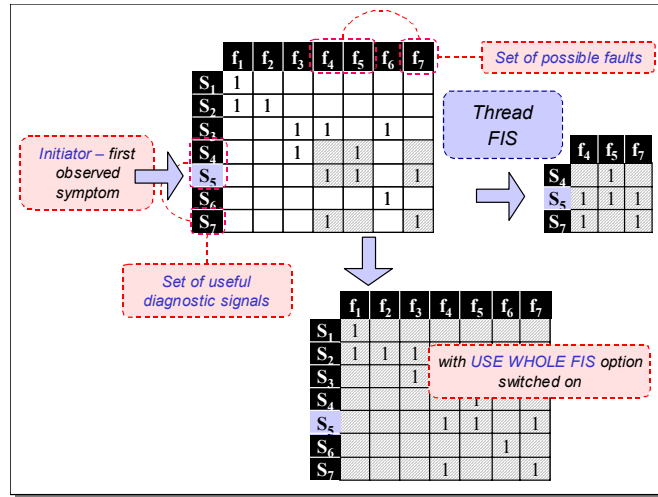


Figure 44. Creating Thread FIS when 'use whole FIS' option is switch on.

Table 12. FDI algorithm parameters of Aggregation category.

Name	FIS level	Values	Description
Simple premise operator	Subprocess	MAX SUM drSUM	Fuzzy OR operator used for simple premise aggregation. See Figure 45. 'drSUM' is a drastic sum operator.
Complex premise operator	Subprocess Global	PROD	Fuzzy AND operator used for complex premise aggregation. See Figure 45.
Rule premise operator	Subprocess	MAX SUM drSUM	Fuzzy OR operator used for rule aggregation. See Figure 45. 'drSUM' is a drastic sum operator.

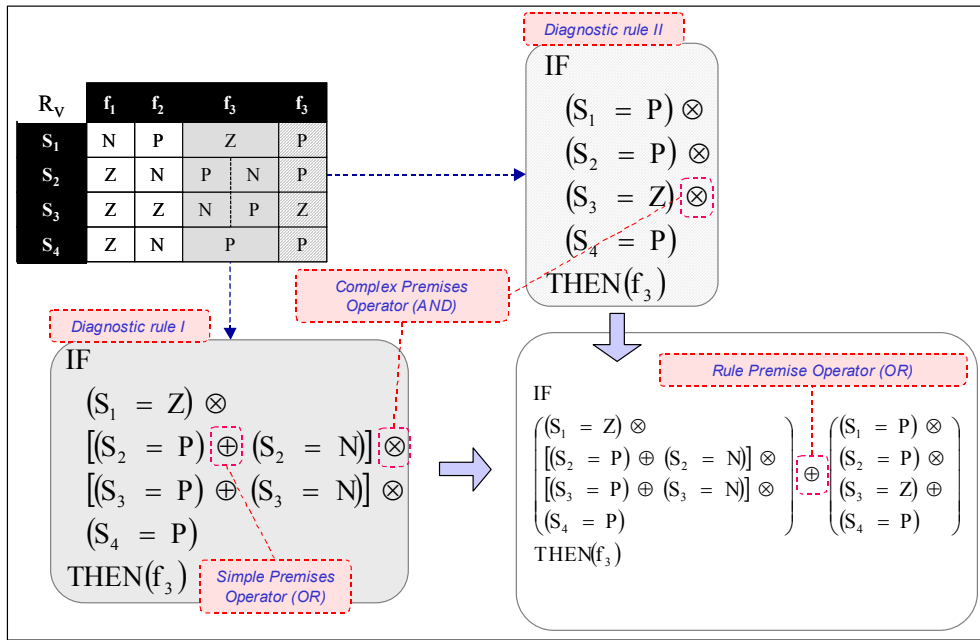
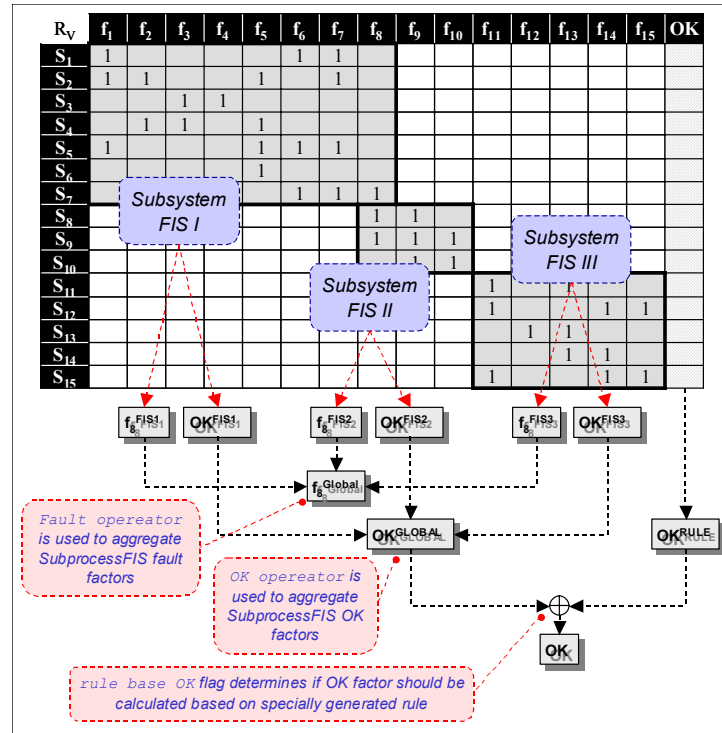


Figure 45. Different operators used during calculating fault certainty factors.

Table 13. FDI algorithm parameters of Detection category.

Name	FIS level	Values	Description
<i>SubFISes aggregation</i>	Global		Determines how fault factors are accommodated on the Global FIS level
<i>Faults operator</i>		MAX PROD	Fault certainty factors calculated on SubFIS level are aggregated with use of ' <i>Fault operator</i> ' on GlobalFIS level. See Figure 46.
<i>OK operator</i>		PROD	Determines the operator for aggregating the certainty factors of the state of complete efficiency calculated on SubFISlevel. See Figure 46.
<i>rule base OK</i>		boolean	If switch on than state of the complete efficiency is calculated based on specially generated rule. If switch off than state of the complete efficiency is calculated based on certainty factors of the state of complete efficiency calculated on SubFISlevel.
<i>normalise</i>	Subprocess		Determines if the calculated fault certainty factors should be normalised so their sum equals 1.

**Figure 46. Accommodate fault certainty factors on GlobalFIS level. Different methods of calculating state of complete efficiency.**